MATH 128A Mid 2 Jie Wen 3041994738

Problem 0: Academic Integrity

I affirm that I have read and will follow all instructions. I will not cheat, and I will not violate academic integrity. As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others. Jie Wen

Problem 1: Nonlinear Equations

(a) Setting $f(x)=0$: $x^2-\frac{2}{x}=0$ Assuming $x\neq 0$: $x^3=2$ ∴ The unique real root: $p=2^{\frac{1}{3}}$

(b) Newton's Method uses the iteration $p_{n+1}=p_n-\frac{f(p_n)}{f'(p_n)}$ ∴ $g(x)=x-\frac{f(x)}{f'(x)}$ ∵ $f(x)=x^2-2x^{-1}$ ∴ $f'(x)=2x+2x^{-2}$

$$\therefore g(x)=x-\frac{x^2-\frac{2}{x}}{2x+\frac{2}{x^2}}=x-\frac{x^4-2x}{2x^3+2}=\frac{(2x^4+2x)-(x^4-2x)}{2x^3+2}=\frac{x^4+4x}{2x^3+2x}$$

(c) *Is the convergence linear?* No.

Justification: ∵ $f(x)=x^2-\frac{2}{x}$, (root) $p=2^{\frac{1}{3}}$, $f(p)=0$ with $p\neq 0$, $f'(x)=2x+2x^{-2}$ ∴ $f'(p)=2p+2p^{-2}$

∵ $p=2^{\frac{1}{3}}>0$ ∴ $f'(p)>0$ ∴ By theorem 2.11, p is a simple zero. Also by theorem 2.8, the convergence is faster than linear.

Is it at least quadratic? Yes.

Justification: According to theorem 2.9, if $g'(p)=0$, the sequence converges at least quadratically.

$$g'(x)=\frac{(4x^3+4)(2x^3+2)-(x^4+4x)(6x^2)}{(2x^3+2)^2}$$

Substitute $x^3=2$: Numerator $=(4\cdot 2+4)(2\cdot 2+2)-(2x+4x)(6x^2)=72-36x^3=0$ ∴ $g'(p)=0$

∴ the convergence is at least quadratic

(d) $f''(x)=2-4x^{-3}$ $f''(p)=0$; $f'''(x)=12x^{-4}$ $f'''(p)=\frac{12}{2^{\frac{4}{3}}}\neq 0$

∵ $f(p)=0$, $f'(p)\neq 0$, $f''(p)=0$, $f'''(p)\neq 0$

∴ The Taylor expansion of the error term will have its first non-zero at the cubic power

∴ The order of convergence is 3

Problem 2: Polynomial Interpolation

(a) By uniqueness of the interpolating polynomial (theorem 3.2) of degree ≤ 5, the minimal-degree polynomial through these 6 points is: $N(x)=x^2$

(b) By uniqueness of the interpolating polynomial (theorem 3.2), his Lagrange polynomial $L(x)$ must simplify to the same expression, $L(x)=x^2$

(c) ∵ For a polynomial of degree n, any divided difference of order $k>n$ is 0. Here, $n=2, k=5$, so $f[0,1,2,3,4,5]=0$

(d) $P(x)=L(x)+f[x_0,x_1,x_2,x_3,x_4,x_5,x_6]\prod_{i=0}^{5}(x-x_i)$ ∴ $P(x)-L(x)=f[0,1,2,3,4,5,\pi]\cdot x(x-1)(x-2)(x-3)(x-4)(x-5)$

(e) ∵ The order of arguments in a divided difference does not matter (constants) ∴ $f[0,1,2,3,\pi,4,5]=f[0,1,2,3,4,5,\pi]:=K$

By (d): $P(\pi)-L(\pi)=K\cdot\pi(\pi-1)(\pi-2)(\pi-3)(\pi-4)(\pi-5)$

By (b): $L(\pi)=\pi^2$, $P(\pi)=3.14^2$ ∴ $K=f[0,1,2,3,\pi,4,5]=\frac{3.14^2-\pi^2}{\pi(\pi-1)(\pi-2)(\pi-3)(\pi-4)(\pi-5)}$

(f) By (e): $P(x)$ is a polynomial of degree 6 with leading coefficient K

∵ $3.14<\pi$ ∴ $3.14^2-\pi^2<0$ ∵ The denominator >0 ∴ $K<0$ ∴ $\lim_{x\to+\infty}P(x)=-\infty$

Problem 3: Numerical Integration

(a) The integration interval: $[a,b]=[-2,4]$ $\quad n=6 \quad \therefore h=\frac{4-(-2)}{6}=1$

Nodes are: $x_0=-2$, $x_1=-1$, $x_2=0$, $x_3=1$, $x_4=2$, $x_5=3$, $x_6=4$

$$\int_{-2}^{4} f(x)\,dx = \sum_{i=0}^{2} (2h) f(x_{2i+1}) = 2\left[f(x_1)+f(x_3)+f(x_5)\right] = 2\left\{\left[h(-1)^3+A\right]+\left[A\right]+\left[g(3)\right]\right\} = 2h(-1)^3+4A+2g(3)$$

(b) On $[-2,0]$: $f(x)=h(x)^3+A$, which is smooth (given) $\Rightarrow$ can apply the midpoint rule error formula

On $[0,2]$: $f(x)=A$ (constant) $\Rightarrow$ The error is 0

On $[2,4]$: $f(x)=g(x)$, which is smooth (given) $\Rightarrow$ can apply the midpoint rule error formula

At $x=0$: $\begin{cases} \lim_{x\to 0^-} f(x)=A \\ \lim_{x\to 0^+} f(x)=A \end{cases} \Rightarrow f$ is continuous at $x=0$

At $x=2$: $\begin{cases} \lim_{x\to 2^-} f(x)=A \\ \lim_{x\to 2^+} f(x)=g(2) \end{cases} \Rightarrow$ For continuity, need $g(2)=A$

Apply Theorem 4.6: For each smooth piece where $f\in C^2$, the midpoint rule error on interval of length $2h$ is:

$$E=\frac{2h\cdot h^2}{6} f''(\mu)=\frac{h^3}{3} f''(\mu) \qquad (\mu_1\in(-2,0),\ \mu_2\in(2,4))$$

$$|E_{total}| \le \frac{h^3}{3}|h''(\mu_1)| + 0 + \frac{h^3}{3}|g''(\mu_2)| \overset{h=1}{=\!=} \frac{1}{3}\left[\max_{x\in[-2,0]}|h''(x)| + \max_{x\in[2,4]}|g''(x)|\right]$$